

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

AI full-stack engineer with 5 years across React/TypeScript, Node/NestJS and AWS. I design and ship production AI agents end to end, with the evaluation and tracing that keep them reliable. AI engineering since 2024.

Skills

AI: LLM agents, tool calling, multi-agent, LangGraph, RAG and vector search (OpenSearch), evaluation, tracing (OpenTelemetry, Langfuse)

Backend: Python, FastAPI, Node.js, NestJS, PostgreSQL

Frontend: React, Next.js, TypeScript

Cloud: AWS, Docker, GitHub Actions

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Stack: LangGraph, Python, TypeScript, NestJS, PostgreSQL, OpenSearch, React, Next.js, AWS

- Designed a configurable AI agent platform (LangGraph, NestJS): one black box became **seven steps**, each testable, on a visual builder over 50 typed functions, so per-customer changes ship without code or a redeploy; running for multiple B2B customers
- Designed the **multi-agent** layer of the WhatsApp sales agent (LangGraph): messages route between service and payment agents, and a guardrail classifier filters spam, auto-replies and off-topic chat before the model answers
- Built the evaluation system for the agent: **64 scenarios** graded line by line against the correct results, run twice daily with Teams alerts; it repeatedly caught breakages from other teams' backend changes before production
- Built the agent's knowledge search (**OpenSearch** vector search, LiteLLM embeddings) with per-customer isolation, so every customer's answers come only from their own documents
- Built a matching engine that maps shorthand user text to catalog items over 15,242 production aliases: the correct first match rose **69–78%** in testing and **81–95%** on real user messages
- Built a multimodal extraction pipeline: photos of handwritten sheets become structured carts through OCR and entity matching, at **88%** line accuracy on a 378-line test set
- Built per-customer agent **configuration**: workspace guidance feeds extraction, matching and OCR prompts, so each customer's terminology works without code changes
- Designed **human-in-the-loop** escalation for the WhatsApp assistant: intents the bot should handle run at once while others wait for a human reply, with per-conversation ordering through message groups
- Built LLM **cost tracking** on a LiteLLM gateway: per-workspace keys and spend tags per feature, so model cost is attributable by customer and feature
- Built end to end in **2.5 weeks**: a WhatsApp workflow (agent, NestJS backend, React web app) that replaced a plant's offline pour and dispatch forms; over 3,800 pour cards and 1,900 dispatches captured across 200 projects since launch
- Built dynamic WhatsApp **template buttons**: five button types resolve at send time into the right page for each buyer, from the template builder to the landing page
- Cut first page load from 15–30 seconds to **1.5 seconds** by removing unused mobile code, rebuilding the Webpack config and lazy loading; builds got 30–40% faster

Sumeru Software – Senior Developer

Aug 2022 – Jan 2023

- Built a role-based CRM from scratch (React, Redux Toolkit, Node.js) with permission-based screens and modular dashboards; owned it from kickoff to production and shipped the MVP in **4 months** with a team of 4

Aerchain – Software Development Engineer

Intern Jul 2019 – Jul 2020; SDE 1 Apr 2021 – Jan 2022

- Built the real-time auction module for a B2B procurement platform, with an **optimistic UI** that reconciles with the server so bids feel instant and stay correct
- Built barcode-scan inventory management as an **offline-first** PWA (service workers) that keeps working without a connection

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020